

Problem

A 50 g cube of ice at -10°C is dropped into 200 g of liquid water at 60°C in a perfectly insulated container (ignore the container's heat capacity and any heat loss to the surroundings).

Given constants:

$$c_{\text{ice}} = 2100 \text{ J/kg}\cdot\text{K}, \quad c_{\text{water}} = 4180 \text{ J/kg}\cdot\text{K}, \quad L_f = 334,000 \text{ J/kg}.$$

Find the final equilibrium temperature T_f .

Solution

Given that,

$$m_i = 0.050 \text{ kg} \quad (\text{ice}), \quad m_w = 0.200 \text{ kg} \quad (\text{water})$$

Heat lost by water

$$Q_{\text{lost}} = m_w c_{\text{water}} (60 - T_f)$$

The ice absorbs heat in three parts:

$$Q_{\text{gained}} = m_i c_{\text{ice}} (10) + m_i L_f + m_i c_{\text{water}} T_f$$

Energy balance Since the system is insulated:

$$m_w c_{\text{water}} (60 - T_f) = m_i c_{\text{ice}} (10) + m_i L_f + m_i c_{\text{water}} T_f$$

Substituting the values

$$(0.200)(4180)(60 - T_f) = (0.050)(2100)(10) + (0.050)(334000) + (0.050)(4180)T_f$$

$$50160 - 836T_f = 1050 + 16700 + 209T_f$$

$$50160 - 17750 = 1045T_f$$

$$32410 = 1045T_f$$

$$T_f \approx \frac{32410}{1045} \approx 31^\circ\text{C}$$

Answer

$$T_f \approx 31^\circ\text{C}$$

Entropy Change in Mixing Ice and Water

Problem

A 50g cube of ice at -10°C is dropped into 200g of liquid water at 60°C inside a perfectly insulated container. Neglect the container's heat capacity and any heat loss to the surroundings.

Given constants:

$$c_{\text{ice}} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}, \quad c_{\text{water}} = 4180 \text{ J kg}^{-1} \text{ K}^{-1}, \quad L_f = 334000 \text{ J kg}^{-1}.$$

Find the **final equilibrium temperature** and the **total entropy change of the system = ice + water**.

Energy balance (final temperature).

Heat absorbed by ice:

$$Q_{\text{ice}} = m_{\text{ice}}c_{\text{ice}}(0 - (-10)) + m_{\text{ice}}L_f + m_{\text{ice}}c_{\text{water}}(T_f - 0).$$

Heat lost by warm water:

$$Q_{\text{water}} = m_{\text{water}}c_{\text{water}}(60 - T_f).$$

Equating:

$$0.050(2100)(10) + 0.050(334000) + 0.050(4180)T_f = 0.200(4180)(60 - T_f).$$

Solving,

$$\boxed{T_f \approx 31.0^\circ\text{C}}$$

Entropy change of ice (three parts).

$$\Delta S_1 = m_{\text{ice}}c_{\text{ice}} \ln \frac{273.15}{263.15} \approx 3.92 \text{ J/K}$$

$$\Delta S_2 = \frac{m_{\text{ice}}L_f}{273.15} \approx 61.14 \text{ J/K}$$

$$\Delta S_3 = m_{\text{ice}}c_{\text{water}} \ln \frac{304.16}{273.15} \approx 22.48 \text{ J/K}$$

Entropy change of water (cooling).

$$\Delta S_{\text{water}} = m_{\text{water}}c_{\text{water}} \ln \frac{304.16}{333.15} \approx -76.10 \text{ J/K}$$

S Total entropy change.

$$\Delta S_{\text{total}} = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_{\text{water}}$$

$$\Delta S_{\text{total}} \approx 3.92 + 61.14 + 22.48 - 76.10$$

$$\boxed{\Delta S_{\text{total}} \approx +11.4 J/K}$$

The positive entropy change confirms that the process is **spontaneous and irreversible**.